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Unit 5 - Knowledge Check

PDF Documents

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KNOWLEDGE CHECK

1 What is the primary function of a solar windmill in renewable energy systems?

- ☐ A To produce chemical energy
- ☐ B To convert solar energy into electrical energy
- ☐ C To reduce solar radiation
- ☐ D To increase humidity levels

CORRECT ANSWER

☒ B

2 Which energy source is commonly used in renewable energy projects like solar windmills?

- ☐ A Coal
- ☐ B Solar
- ☐ C Natural Gas
- ☐ D Petroleum

CORRECT ANSWER

☒ B

3 What impact does using renewable energy have on the environment?

- ☐ A Increases air pollution
- ☐ B Reduces carbon emissions
- ☐ C Causes water pollution
- ☐ D All of the above

CORRECT ANSWER

☒ B

4 Why is humidity measurement important in environmental monitoring for renewable energy sources?

- ☐ A It affects the comfort of workers
- ☐ B It influences the efficiency of solar panels
- ☐ C It changes the color of the equipment
- ☐ D It increases energy consumption

CORRECT ANSWER

☒ B

5 What is the purpose of using an 'if else' statement in programming a solar windmill?

- ☐ A To choose between two energy sources
- ☐ B To control the display messages based on energy levels
- ☐ C To increase the speed of the windmill
- ☐ D To connect to the internet

CORRECT ANSWER

☒ B

6 What happens if you change an 'and' operator to an 'or' operator in a condition checking both energy sufficiency and motor status?

- ☐ A The system becomes less secure
- ☐ B It will display 'Off' more frequently
- ☐ C The system will display messages only when both conditions are met
- ☐ D The condition becomes true if either or both situations are true

CORRECT ANSWER

☒ D

7 Why would you use variables in programming the windmill?

- ☐ A To store temporary data like energy levels
- ☐ B To make the program longer
- ☐ C To create graphics for the display
- ☐ D To connect with other windmills

CORRECT ANSWER

☒ A



Critical Thinking Challenge

- 1 Discuss how the integration of solar and wind power in a single renewable energy project like a solar windmill could benefit your local community economically and environmentally.

- 2 Reflect on how changing one programming variable in the solar windmill could significantly alter its functionality. Provide an example of such a change and its potential impacts.